Question 1

Assume the following instruction trace. ADD/SUB operations take one cycle, MUL operations take three cycles, and Load/Store operations take two cycles. The pipeline has 2 ADD/SUB units, 1 MUL unit, and 1 Load/Store unit. The functional units are pipelined.

A. add R1, R2, R2

B. mul R3, R1, R3

C. lw R2, 0(R1)

D. sub R4, R3, R2

E. mul R6, R5, R5

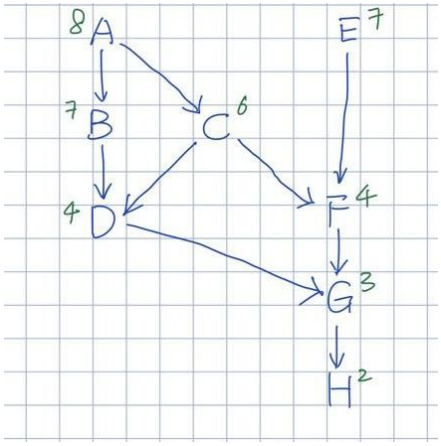
F. add R7, R6, R2

G. add R7, R4, R7

H. sw R7, 0(R8)

a) Draw the dependence graph for the 8 instructions shown above. Include all anti, output, and pure dependencies with arcs.

b) Show the dependence height of each instruction (multiple answers possible).



INST	height
A	8
B	7
E	7
C	6
D	4
F	4
G	3
H	2

c) Show the best VLIW schedule you can find. You may show the schedule using letters only.

ALU 1				ALU 2				MUL				LOAD / STORE		
INST	DEST	SRC1	SRC2	INST	DEST	SRC1	SRC2	INST	DEST	SRC1	SRC2	INST	D / S	SRC
A	R1	R2	R2					E	R6	R5	R5			
								B	R3	R1	R3	C	R2	R1
D	R4	R3	R2											
F	R7	R6	R2											
G	R7	R4	R7											
												H	R7	R8

d) Based on your result from c), please generate an encoding scheme for the instructions using the tail and pause bits scheme.

INST	OP	DEST	SRC 1	SRC 2	TAIL	PAUSE
A	ADD	R1	R2	R2	0	0
E	MUL	R6	R5	R5	1	0
B	MUL	R3	R1	R3	0	0
C	LW	R2	R1	-	1	0
D	SUB	R4	R6	R2	1	1
F	ADD	R7	R6	R2	1	0
G	ADD	R7	R4	R7	1	0
H	SW	R7	R8	-	1	0

Question 2

Consider the following loop. (Although it's not relevant to the solution, you can assume that R3 begins with a negative value stored in it.)

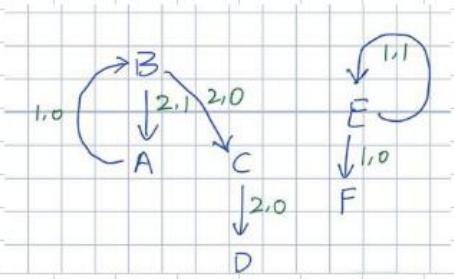
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LABEL:A: ADD R1, R1, #1
      B: MUL R1, R1, 4
      C: LW   R2, 100(R1)
      D: SW   R2, 200(R1) // Note: R2 is a SOURCE register for SW
      E: ADD R3, R3, #1
      F: BRNEZ R3, LABEL // branch to LABEL if R3 != 0
```

Referring to Software Pipelining via Iterative Modulo Scheduling, what is the RecMII and ResMII for this loop? What is the overall minimum II? Assume there is 1 ALU unit, 1 MUL unit, 1 LW unit and 1 SW unit. Both MUL and LW takes **2** cycles and are pipelined. SW, ADD and BREZ all take 1 cycle. ADD and BREZ use the ALU unit. Now show a prologue, kernel and epilog after modulo scheduling. For simplicity, you can write the letter of the operation in a table. You do not need to write the instructions themselves.

RRB based instruction stream

ADD R1[0], R1[1], #1
MUL R1[0], R1[0], 4
LW R2[0], 100(R1[0])
SW R2[0], 200(R1[0])
ADD R3[0], R3[1], #1
BNE R3[0], LABEL

Cycle Dependency Graph



	ALU	MUL	LW	SW
FU used	2	1	1	1
ops	3	1	1	1
quotient	2	1	1	1
ResMII = max(2, 1, 1, 1) = 2				

cycle	cycle length	iteration dist	quotient
BAB	3	1	3
EE	1	1	1
RecMII = max(3, 1) = 3			

Overall MinII = max(ResMII, RecMII) = 3

Modulo Reservation Table

ALU	MUL	LW	SW
A		C	
E	B		
F			D

Schedule

ALU	MUL	LW	SW	
A				**Prolog**
E	B			
F*				
A		C		**Kernel**
E	B			
F**			D	
		C		**Epilog**
			D	